CS401: Computer Architecture & Assembly Language Programming

Premium Final Term Conceptual Study Guide

Author: ZB ZAIB

Contact: +92 300 5427579 / 03435583148

Quranic Motivation (Roman English)

> “Insaan ke liye wahi kuch hota hai jis ke liye woh khud mehnat aur koshish karta hai.” **Reference:** Surah An-Najm, Ayah 39

File Introduction

- This file follows the **Final Term syllabus only** (Topic 9.1, Page 105 to 184).
- Lessons are **deeply explained** to ensure conceptual clarity.
- Definitions are written in **very simple English** for easy understanding.
- **Daily life examples** are used to explain technical assembly concepts.
- VU exams are **conceptual**; focus on "How" and "Why" rather than just "What".
- Students must read **LMS handouts at least three times**.
- This file **supports LMS handouts**; it does not replace them.

Lesson-wise Content (In the Name of Allah)

Topic 9.1: Hardware Interrupts

Concept:

Hardware interrupts are signals sent by external devices (like keyboard, mouse, or timer) to the processor. When a device needs attention, it "interrupts" the processor's current work. The processor stops, handles the request, and then goes back to its original task.

Deep Explanation:

The processor is always busy executing instructions. If a key is pressed, the keyboard controller sends a signal to the **Programmable Interrupt Controller (PIC)**. The PIC then

tells the processor that a device needs help. The processor uses an **Interrupt Vector Table (IVT)** to find the address of the code (ISR) that knows how to handle that specific device.

> **Example:**
>
> Imagine you are reading a book (Processor executing a program). Suddenly, your doorbell rings (Hardware Interrupt). You put a bookmark in your book (Save current state), go to the door to check who it is (Execute ISR), and after talking to the person, you come back and start reading from where you left off (Resume program).

Topic 9.2: I/O Ports

Concept:

I/O ports are like "mailboxes" or "doors" through which the processor talks to hardware devices. Every device has a specific port number (address).

Deep Explanation:

Unlike memory which stores data, I/O ports are used to control hardware. The processor uses special instructions like `IN` (to read from a device) and `OUT` (to send data to a device). In the IBM PC, there are 65,536 possible I/O ports.

> **Example:**
>
> Think of a large apartment building. Each apartment has a mailbox (I/O Port). If the manager (Processor) wants to give a message to a specific resident (Hardware Device), he puts it in that specific mailbox. If he wants to receive a message, he checks that mailbox.

Topic 9.3: Terminate and Stay Resident (TSR)

Concept:

A TSR is a special type of program that, after it finishes its initial execution, stays in the computer's memory (RAM) and continues to work in the background.

Deep Explanation:

Normally, when a program ends, its memory is freed for other programs. But a TSR uses a special DOS service (`INT 21h, Service 31h`) to tell the operating system: "I am done for now, but keep my code in memory because I need to respond to interrupts later."

> **Example:**
>
> Think of a security guard at a bank. He arrives in the morning (Program starts), sets up his chair (Initialization), and then just sits there (Stays Resident). He isn't actively "working" until someone tries to enter (Interrupt), then he acts, and goes back to sitting.

Topic 10: Video Services & BIOS/DOS Interrupts

Concept:

Instead of writing complex code to control the screen or hardware directly, we use "Services" provided by BIOS (Basic Input/Output System) and DOS.

Deep Explanation:

BIOS interrupts (like `INT 10h` for video) are built into the hardware's ROM. DOS interrupts (like `INT 21h`) are provided by the Operating System. These services make a programmer's life easy because they handle the difficult hardware details for us.

> **Example:**
>
> If you want to travel to another city, you don't build a car yourself (Direct Hardware Access). Instead, you call a taxi service (BIOS/DOS Interrupt). You just tell the driver where to go, and he handles the engine, steering, and roads for you.

Topic 11: Serial and Parallel Ports

Concept:

These are ways to connect external devices like printers (Parallel) or modems (Serial) to the computer.

Deep Explanation:

- **Parallel Port:** Sends multiple bits of data at the same time (like an 8-lane highway). It is fast but only for short distances.
- **Serial Port:** Sends data one bit at a time (like a single-lane road). It is slower but can work over long distances.

> **Example:**
>
> **Parallel:** 8 people walking side-by-side through a wide gate.
>
> **Serial:** 8 people walking one after another through a narrow door.

Conceptual Questions (Short & Long)

Q1: Why do we push all registers at the start of an ISR?

Answer:

We push registers to save their current values. Since an interrupt can happen at any time, the registers might be holding important data for the main program. If the ISR changes them, the main program will crash when it resumes.

Step-by-step:

1. Interrupt occurs.
2. ISR starts.
3. `PUSH` all registers to the Stack (Saving).
4. ISR does its work.
5. `POP` all registers from the Stack (Restoring).
6. `IRET` to go back.

> **Example:** Before you let a guest use your desk, you take all your important papers and put them in a drawer (Push). After the guest leaves, you take your papers back out (Pop) so you can continue your work.

Q2: What is the difference between `INT` and `CALL`?

Answer:

`CALL` is used to run a subroutine within your own program. `INT` is used to trigger an interrupt, which often involves the hardware or the operating system. `INT` also automatically saves the "Flags Register" on the stack, while `CALL` does not.

> **Example:** `CALL` is like calling your brother in the next room to help you. `INT` is like calling the police (External Service) for help.

Lesson Quiz (Conceptual MCQs)

1. **Which instruction is used to return from an Interrupt Service Routine?**
   - A) RET
   - B) IRET (Correct)
   - C) EXIT
   - *Reasoning:* `IRET` is special because it restores the Flags register along with the instruction pointer.
   - *Example:* Like a "Resume" button that not only starts the video but also restores the volume and brightness you had before.
2. **What is the size of one entry in the Interrupt Vector Table (IVT)?**
   - A) 2 bytes
   - B) 4 bytes (Correct)
   - C) 8 bytes

- *Reasoning:* Each entry stores a 2-byte Offset and a 2-byte Segment address (Total 4 bytes).
- *Example:* Like a phone book entry that has both a Name and a Number.

3. **Which port is typically used for the Keyboard in an IBM PC?**
   - A) Port 60h (Correct)
   - B) Port 378h
   - C) Port 20h
   - *Reasoning:* Port 60h is the standard data port for the keyboard controller.
4. **What does EOI stand for in the context of PIC?**
   - A) End of Instruction
   - B) End of Interrupt (Correct)
   - C) Every Other Interrupt
   - *Reasoning:* It tells the PIC that the current interrupt is finished so it can send the next one.
5. **Which DOS service is used for TSR programs?**
   - A) INT 21h, Service 4Ch
   - B) INT 21h, Service 31h (Correct)
   - C) INT 10h, Service 00h
   - *Reasoning:* Service 31h terminates the program but keeps it in memory.

Practice Final Term Paper

Section 1: Quizzes (30 MCQs - Sample)

(Note: 100+ Quizzes are provided in the final section)

Section 2: Short Questions (5 Questions)

Q1: Define IRQ.

Answer: IRQ stands for Interrupt Request. It is a physical wire or signal line that a hardware device uses to tell the PIC it needs attention.

> **Example:** Like a student raising their hand in class to ask a question.

Q2: What is the purpose of the Programmable Interrupt Controller (PIC)?

Answer: The PIC acts as a manager. It receives multiple interrupt requests from different devices, decides which one is more important (Priority), and sends them one by one to the processor.

> **Example:** A receptionist in a doctor's office who decides which patient goes in first based on their appointment or urgency.

(...Additional 3 short questions follow the same conceptual pattern...)

Section 3: Long Questions (3 Questions)

Q1: Explain the complete process of how a key press is handled by the system.

Answer:

1. User presses a key.
2. Keyboard controller generates an IRQ 1 signal.
3. PIC receives IRQ 1 and signals the Processor.
4. Processor finishes current instruction and asks PIC for the interrupt number.
5. PIC sends number 9 (for IRQ 1).
6. Processor looks at IVT entry 9 (Address 9*4 = 36).
7. Processor jumps to the Keyboard ISR.
8. ISR reads the scan code from Port 60h.
9. ISR sends EOI to PIC.
10. Processor returns to the original program using `IRET`.

> **Example:** Like a fire alarm going off. The sensor detects smoke (Key press), sends a signal to the main panel (PIC), the panel rings the bell (Processor), firefighters arrive (ISR), they put out the fire (Read scan code), reset the alarm (EOI), and everyone goes back to sleep (IRET).

Important Definitions (Syllabus 105-184)

Term	Simple Definition
ISR	A small program that handles a specific interrupt.
IVT	A table in memory that holds the addresses of all ISRs.

Scan Code	A unique number sent by the keyboard for every key press or release.
Hooking	Replacing a standard ISR address in the IVT with your own custom ISR address.
Polling	When the processor constantly checks a device to see if it's ready (Opposite of Interrupt).

Final Quiz Bank (100+ Conceptual Items)

(Detailed list of 100+ MCQs with reasoning and examples follows here...)

Guidance & Motivational Advice

- **VU exams are conceptual:** Don't just memorize the code; understand why we use `push`, `pop`, and `int`.
- **Only LMS handouts are reliable:** Many WhatsApp files have wrong answers. Stick to the official handouts.
- **Focus on understanding:** If you understand the "Doorbell" example, you understand Hardware Interrupts.
- **Motivational Advice:** Success is not final, failure is not fatal: it is the courage to continue that counts. Your hard work today will define your 4.0 CGPA tomorrow.

Author: ZB ZAIB

Contact: +92 300 5427579 / 03435583148

Final Quiz Bank (100+ Conceptual Items)

1. **What is the primary purpose of an Interrupt?**
    - **Answer:** To allow external devices to get the processor's attention.
    - **Reasoning:** Without interrupts, the processor would have to waste time checking every device (polling).
    - **Example:** Like a phone ringing instead of you checking the screen every 5 seconds to see if someone called.
2. **Where is the Interrupt Vector Table (IVT) located in memory?**
    - **Answer:** At the very beginning of memory (address 0000:0000).

- **Reasoning:** It is a fixed location so the processor always knows where to find it.

3. **How many interrupts can the IVT hold?**
   - **Answer:** 256 interrupts.
   - **Reasoning:** The interrupt number is 8 bits (0 to 255).
4. **What is the size of the IVT in bytes?**
   - **Answer:** 1024 bytes (256 interrupts * 4 bytes each).
5. **Which interrupt is used for the System Timer?**
   - **Answer:** INT 8.
   - **Reasoning:** The timer hardware is connected to IRQ 0, which maps to INT 8.
6. **How many times per second does the timer interrupt occur?**
   - **Answer:** 18.2 times per second.
   - **Reasoning:** This is the standard frequency of the IBM PC timer.
7. **What is the purpose of the `STI` instruction?**
   - **Answer:** Set Interrupt Flag (Enable interrupts).
   - **Reasoning:** It allows the processor to respond to maskable hardware interrupts.
8. **What is the purpose of the `CLI` instruction?**
   - **Answer:** Clear Interrupt Flag (Disable interrupts).
   - **Reasoning:** Used to prevent interrupts during critical code sections.
9. **Which register is automatically saved on the stack during an `INT` call but not a `CALL`?**
   - **Answer:** Flags Register.
10. **What is a 'Software Interrupt'?**
    - **Answer:** An interrupt triggered by the program code using the `INT` instruction.
    - **Example:** Requesting DOS to print a string.
11. **What is a 'Hardware Interrupt'?**
    - **Answer:** An interrupt triggered by a physical device.
    - **Example:** Moving the mouse.
12. **What is an 'Exception'?**
    - **Answer:** An internal interrupt caused by an error in the code.
    - **Example:** Dividing a number by zero.
13. **Which IRQ is assigned to the Keyboard?**

- **Answer:** IRQ 1.

14. **Which IRQ is assigned to the Serial Port (COM1)?**
    - **Answer:** IRQ 4.
15. **Which IRQ is assigned to the Parallel Port (LPT1)?**
    - **Answer:** IRQ 7.
16. **What is the I/O port address for the PIC's Command register?**
    - **Answer:** 20h.
17. **What is the I/O port address for the PIC's Data register (IMR)?**
    - **Answer:** 21h.
18. **What does IMR stand for?**
    - **Answer:** Interrupt Mask Register.
    - **Reasoning:** It is used to enable or disable specific IRQ lines.
19. **If a bit in the IMR is set to 1, what does it mean?**
    - **Answer:** The corresponding IRQ is masked (disabled).
20. **What is the 'Master' and 'Slave' PIC configuration?**
    - **Answer:** Two PICs are connected to handle 15 IRQs (8 from Master, 7 from Slave).
21. **Which IRQ on the Master PIC is used to connect the Slave PIC?**
    - **Answer:** IRQ 2.
22. **What is the purpose of the `OUT` instruction?**
    - **Answer:** To send data from a register to an I/O port.
23. **What is the purpose of the `IN` instruction?**
    - **Answer:** To read data from an I/O port into a register.
24. **Which register must be used for the port address in `IN` / `OUT` if the address is greater than 255?**
    - **Answer:** DX register.
25. **What is 'Memory Mapped I/O'?**
    - **Answer:** When I/O devices are accessed as if they were memory locations.
26. **What is 'Isolated I/O'?**
    - **Answer:** When I/O devices have their own separate address space (used by Intel).
27. **What is the video memory address for Text Mode?**
    - **Answer:** 0xB8000.

28. **How many bytes are used for one character in text mode?**
    - **Answer:** 2 bytes (1 for ASCII, 1 for Attribute).
29. **What does the 'Attribute Byte' control?**
    - **Answer:** Foreground color, background color, and blinking.
30. **Which BIOS interrupt handles Video Services?**
    - **Answer:** INT 10h.
31. **Which DOS service is used to print a '$' terminated string?**
    - **Answer:** INT 21h, Service 9.
32. **Which register holds the address of the string for DOS Service 9?**
    - **Answer:** DX (Offset) and DS (Segment).
33. **What is the scan code?**
    - **Answer:** A unique 1-byte code sent by the keyboard for each key.
34. **What is a 'Make Code'?**
    - **Answer:** The scan code sent when a key is pressed.
35. **What is a 'Break Code'?**
    - **Answer:** The scan code sent when a key is released (Make code + 0x80).
36. **Which port is used to read the keyboard scan code?**
    - **Answer:** Port 60h.
37. **What is the purpose of Port 61h?**
    - **Answer:** Keyboard Control/Acknowledgment port.
38. **What is a 'TSR' program?**
    - **Answer:** Terminate and Stay Resident.
39. **Which interrupt is used to make a program TSR?**
    - **Answer:** INT 21h, Service 31h.
40. **In a TSR, what does the 'Resident' part do?**
    - **Answer:** It stays in memory to handle future interrupts.
41. **In a TSR, what does the 'Transient' part do?**
    - **Answer:** It initializes the program and then is discarded.
42. **What is 'Interrupt Hooking'?**
    - **Answer:** Redirecting an interrupt to your own code by changing its IVT entry.

43. **Why must we save the old ISR address when hooking?**
    - **Answer:** To call the original handler or restore it when our program ends.
44. **What is the 'Chain' technique in interrupts?**
    - **Answer:** Calling the old ISR from within your new ISR.
45. **What is the Parallel Port's base address?**
    - **Answer:** 378h (for LPT1).
46. **How many data bits does the Parallel Port send at once?**
    - **Answer:** 8 bits.
47. **What is the Serial Port's base address?**
    - **Answer:** 3F8h (for COM1).
48. **What is 'Baud Rate'?**
    - **Answer:** The speed of data transmission in serial communication.
49. **What is 'Parity' in serial communication?**
    - **Answer:** A simple error-checking bit.
50. **What is 'Start Bit' and 'Stop Bit'?**
    - **Answer:** Bits used to mark the beginning and end of a byte in serial data.
51. **Which BIOS interrupt handles Serial Port services?**
    - **Answer:** INT 14h.
52. **Which BIOS interrupt handles Parallel Port services?**
    - **Answer:** INT 17h.
53. **What is 'Polling'?**
    - **Answer:** The processor repeatedly checking a status bit of a device.
54. **Why is Polling inefficient?**
    - **Answer:** It wastes processor cycles waiting for slow devices.
55. **What is 'Interrupt Driven I/O'?**
    - **Answer:** The device signals the processor only when it is ready.
56. **What is the 'Programmable Interval Timer' (PIT)?**
    - **Answer:** The chip (8253/8254) that generates the system timer ticks.
57. **What is the I/O port for PIT Channel 0?**
    - **Answer:** 40h.

58. **What is the I/O port for PIT Control Word?**
    - **Answer:** 43h.
59. **What is 'Real Mode'?**
    - **Answer:** The 16-bit operating mode of Intel processors (used in DOS).
60. **What is 'Protected Mode'?**
    - **Answer:** A 32-bit mode with memory protection and virtual memory.
61. **Which register points to the Global Descriptor Table (GDT)?**
    - **Answer:** GDTR.
62. **What is a 'Selector' in Protected Mode?**
    - **Answer:** A value loaded into a segment register to select a descriptor from the GDT.
63. **What is a 'Descriptor'?**
    - **Answer:** An 8-byte entry in the GDT that describes a memory segment.
64. **What is 'VGA'?**
    - **Answer:** Video Graphics Array.
65. **What is the resolution of standard VGA Mode 13h?**
    - **Answer:** 320x200 pixels with 256 colors.
66. **What is the memory address for VGA Mode 13h?**
    - **Answer:** 0xA0000.
67. **How many bytes per pixel in Mode 13h?**
    - **Answer:** 1 byte.
68. **What is the 'DAC' in VGA?**
    - **Answer:** Digital-to-Analog Converter (handles colors).
69. **What is 'Vertical Retrace'?**
    - **Answer:** The time when the electron beam moves from the bottom to the top of the screen.
70. **Why do we wait for Vertical Retrace?**
    - **Answer:** To avoid "flicker" when updating graphics.
71. **Which port is used to check the VGA status (including retrace)?**
    - **Answer:** 3DAh.
72. **What is 'Double Buffering'?**
    - **Answer:** Drawing on a hidden screen and then copying it to the visible screen.

73. **What is 'DMA'?**
    - **Answer:** Direct Memory Access.
74. **What is the purpose of DMA?**
    - **Answer:** To transfer data between memory and devices without using the processor.
75. **Who is the 'Master' during a DMA transfer?**
    - **Answer:** The DMA Controller.
76. **What is 'Cycle Stealing' in DMA?**
    - **Answer:** When the DMA controller takes control of the bus for a short time.
77. **What is 'Burst Mode' in DMA?**
    - **Answer:** When the DMA controller transfers a large block of data at once.
78. **Which IRQ is used by the Floppy Disk Controller?**
    - **Answer:** IRQ 6.
79. **Which IRQ is used by the Hard Disk Controller?**
    - **Answer:** IRQ 14.
80. **What is 'NMI'?**
    - **Answer:** Non-Maskable Interrupt.
81. **Can NMI be disabled using `CLI`?**
    - **Answer:** No.
82. **What is the purpose of NMI?**
    - **Answer:** For critical hardware failures (like memory parity error).
83. **What is 'Interrupt Latency'?**
    - **Answer:** The time delay between an interrupt request and the start of the ISR.
84. **What is 'Reentrant Code'?**
    - **Answer:** Code that can be safely interrupted and called again.
85. **Is DOS reentrant?**
    - **Answer:** No (this is why TSRs must be careful when calling DOS).
86. **What is the 'In-DOS' flag?**
    - **Answer:** A flag used by TSRs to check if DOS is currently busy.
87. **What is 'Critical Section'?**
    - **Answer:** A part of the code that must not be interrupted.

88. **What is 'Atomic Operation'?**
    - **Answer:** An operation that completes fully without being interrupted.
89. **What is 'Handshaking' in communication?**
    - **Answer:** Signals used to coordinate data transfer between two devices.
90. **What is 'RS-232'?**
    - **Answer:** A standard for serial communication.
91. **What is 'ASCII'?**
    - **Answer:** American Standard Code for Information Interchange.
92. **What is the ASCII code for 'Enter' key?**
    - **Answer:** 13 (0Dh).
93. **What is the ASCII code for 'Space' key?**
    - **Answer:** 32 (20h).
94. **What is 'Little Endian'?**
    - **Answer:** Storing the least significant byte at the lowest address (used by Intel).
95. **What is 'Big Endian'?**
    - **Answer:** Storing the most significant byte at the lowest address.
96. **What is 'Stack Overflow'?**
    - **Answer:** When the stack grows beyond its allocated memory.
97. **What is 'Segment Wrap-around'?**
    - **Answer:** When an offset exceeds 0xFFFF and goes back to 0x0000.
98. **What is 'A20 Line'?**
    - **Answer:** The 21st address line used to access memory above 1MB.
99. **What is 'HMA'?**
    - **Answer:** High Memory Area (the first 64KB above 1MB).
100. **What is 'UMB'?**
    - **Answer:** Upper Memory Blocks (memory between 640KB and 1MB).
101. **What is the ultimate goal of studying CS401?**
    - **Answer:** To understand how software controls hardware at the lowest level.
    - **Advice:** Stay focused, practice coding, and you will achieve a 4.0 CGPA!

END OF FILE